

Online Appendix for:
Unexpected Online Financial News and Intraday
Market Reactions:
An Event-Study of EUR/USD and Nasdaq-100 Prices

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This document collects auxiliary tables and material referenced in the main paper but not included in the main text. Section and table numbering follows the OA prefix and is independent of the main paper. Bibliographic references follow the same convention; the bibliography file is shared between this Online Appendix and the main paper.

A Robustness Tables

This appendix collects auxiliary tables referenced in the main body. All quantities are computed from the same pipeline and seed documented in Section A.

A.1 Cluster-gap-threshold robustness for H1

Section 5.3 treats the 15-minute cluster-gap threshold as a design parameter and notes that the central H1 result is insensitive to the choice within the range 5–30 minutes. Table OA1 reports the H1 event/baseline magnitude ratio for the four thresholds. The ratio is monotone in the threshold (tighter thresholds retain more independent clusters and yield slightly higher ratios), but remains above 1.0 in every cell, the central conclusion in Section 6.1.

A.2 Subsample analysis by LLM sentiment label

The bull-versus-bear asymmetry test (H12) reported as null in Section 8 aggregates over all events. Tables OA2 and OA3 report the magnitude and direction results stratified by

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Table OA1: Cluster-gap-threshold robustness for the H1 magnitude ratio.

Asset	Window	Cluster gap threshold			
		5 min	10 min	15 min	30 min
EUR/USD	+5m	1.90×	1.85×	1.76×	1.67×
EUR/USD	+15m	1.83×	1.77×	1.69×	1.60×
EUR/USD	+60m	1.66×	1.62×	1.58×	1.50×
NDX	+5m	1.73×	1.65×	1.56×	1.48×
NDX	+15m	1.71×	1.66×	1.57×	1.47×
NDX	+60m	1.57×	1.51×	1.43×	1.38×
<i>Number of clusters after dedupe</i>					
EUR/USD +5m		1,557	1,395	1,288	1,069
EUR/USD +15m		1,539	1,379	1,273	1,057
EUR/USD +60m		1,469	1,313	1,208	1,001
NDX +5m		1,457	1,305	1,200	993
NDX +15m		1,435	1,283	1,178	972
NDX +60m		1,350	1,200	1,099	910

Notes. H1 event/baseline magnitude ratio at four cluster-gap thresholds. The 15-minute column corresponds to the specification used in the main body. Tighter thresholds (5–10 min) retain more independent clusters and yield slightly higher ratios; a wider threshold (30 min) compresses the sample but the central finding survives in every cell.

the LLM sentiment label (bull, bear, neutral). The magnitude effect is broadly symmetric across the three subsamples and confirms the H12 null finding at a more granular level. The direction hit rate by sentiment label shows one notable cell: EUR/USD events labelled bull achieve a 57.3% hit rate at the +60-minute window ($p = 0.0007$ on a one-sided binomial); the corresponding bear subsample is not statistically distinguishable from chance. We report this observation as exploratory; it does not feature in the main results because the same conclusion would not hold under a stricter joint-testing correction across the eighteen subsample cells reported here.

A.3 LLM prompt full text

The production LLM call uses the system prompt below. The benchmark-adapted prompt for Financial PhraseBank (Section 5.9) is identical to this prompt up to the schema (positive/negative/neutral in place of the USD/NDX bull/bear/neutral fields) and the omission of the asset- specific instruction.

You are a financial sentiment analyst. Classify how a news event likely affects the US Dollar (USD) and the Nasdaq-100 index (NDX) in the very short term (minutes to hours after publication). Output STRICTLY a single JSON object with this exact schema: sentiment_usd $\in \{\text{bull, bear, neutral}\}$, sentiment_ndx in the same support, signed directional_strength_usd and directional_strength_ndx in $[-1, +1]$,

Table OA2: Magnitude by sentiment label (subsample analysis).

Asset	Window	Sentiment	N	Mean $ r $ (%)	Median $ r $ (%)
EUR/USD	+5m	bull	557	0.0333	0.0207
EUR/USD	+5m	bear	464	0.0330	0.0207
EUR/USD	+5m	neutral	384	0.0306	0.0188
EUR/USD	+15m	bull	557	0.0547	0.0346
EUR/USD	+15m	bear	464	0.0527	0.0360
EUR/USD	+15m	neutral	384	0.0518	0.0311
EUR/USD	+60m	bull	557	0.1006	0.0677
EUR/USD	+60m	bear	464	0.1025	0.0714
EUR/USD	+60m	neutral	384	0.0935	0.0630
NDX	+5m	bull	475	0.0816	0.0506
NDX	+5m	bear	649	0.0923	0.0554
NDX	+5m	neutral	281	0.0700	0.0458
NDX	+15m	bull	475	0.1409	0.0933
NDX	+15m	bear	649	0.1471	0.0931
NDX	+15m	neutral	281	0.1430	0.0866
NDX	+60m	bull	475	0.2619	0.1606
NDX	+60m	bear	649	0.2806	0.1745
NDX	+60m	neutral	281	0.2539	0.1751

Notes. Event-window absolute return $|r|$ stratified by LLM sentiment label. The magnitude effect is broadly symmetric across bull, bear, and neutral subsamples; the small differences do not motivate a separate H12-style asymmetry claim.

Table OA3: Direction hit rate by LLM sentiment label.

Asset	Window	Sentiment	N	Hit rate	CI95	p -binom
EUR/USD	+5m	bull	523	50.1%	[0.458, 0.544]	0.500
EUR/USD	+5m	bear	475	54.9%**	[0.505, 0.594]	0.017
EUR/USD	+15m	bull	521	52.4%	[0.481, 0.567]	0.147
EUR/USD	+15m	bear	470	52.6%	[0.480, 0.570]	0.144
EUR/USD	+60m	bull	494	57.3%***	[0.529, 0.616]	0.0007
EUR/USD	+60m	bear	450	50.4%	[0.458, 0.550]	0.444
NDX	+5m	bull	441	54.0%*	[0.493, 0.586]	0.053
NDX	+5m	bear	560	51.8%	[0.476, 0.559]	0.211
NDX	+15m	bull	434	53.5%*	[0.488, 0.581]	0.082
NDX	+15m	bear	551	49.7%	[0.456, 0.539]	0.568
NDX	+60m	bull	403	53.8%*	[0.490, 0.587]	0.067
NDX	+60m	bear	515	46.2%	[0.420, 0.505]	0.961

Notes. Direction hit rate separately for events labelled bull and bear by the LLM (neutral events excluded as non-directional). Bull-labelled EUR/USD events at +60 minutes reach 57.3% hit rate ($p = 0.0007$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ on raw p -values.

expected_magnitude in {low, med, high}, surprise_level in {expected, surprise, shock}, confidence in [0,1], and a short rationale.

Rules: base judgment strictly on the supplied text; do not invent context; if ambiguous or off-topic, return neutral with low confidence (<0.4); USD and NDX are different assets and their sentiments can differ (e.g., risk-off events: bull USD, bear NDX); directional_strength values respect sentiment direction; return ONLY the JSON object, no preamble.

Five few-shot examples covering geopolitical escalation, dovish Fed signal, hawkish ECB signal, neutral aid announcement, and corporate earnings miss anchor the format.

References